



CPSC 100

Computational Thinking

Data Representation

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University of British Columbia



Agenda

- Course Admin
- Learning Goals
- Practicing Colour conversion
- RGB Colours

Course Admin



Learning Goals

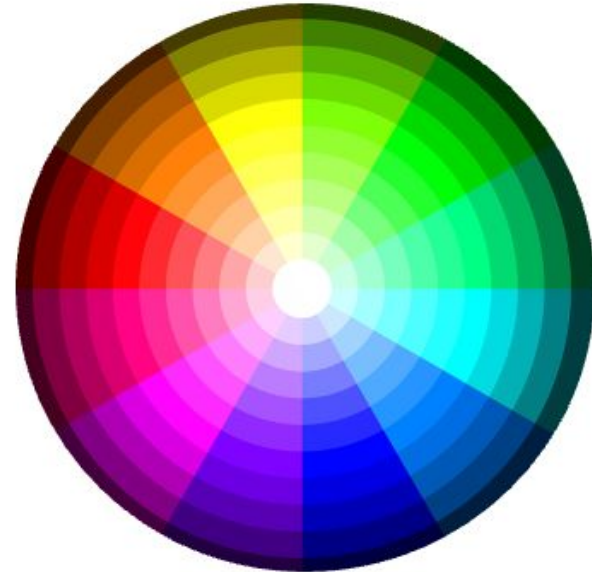
After this **today's lecture**, you should be able to:

- Explain what **ASCII** and **Unicode** are, including their historical context, purpose, and significance in computing.
- **Decode** an **ASCII** representation of a short text document
 - (with a list of ASCII codes provided)
- Articulate **why Unicode was created** and how it solved the problems of earlier encoding systems like ASCII
- **Compare and contrast Unicode with ASCII** in terms of character range, encoding length, and use cases

Learning Goals

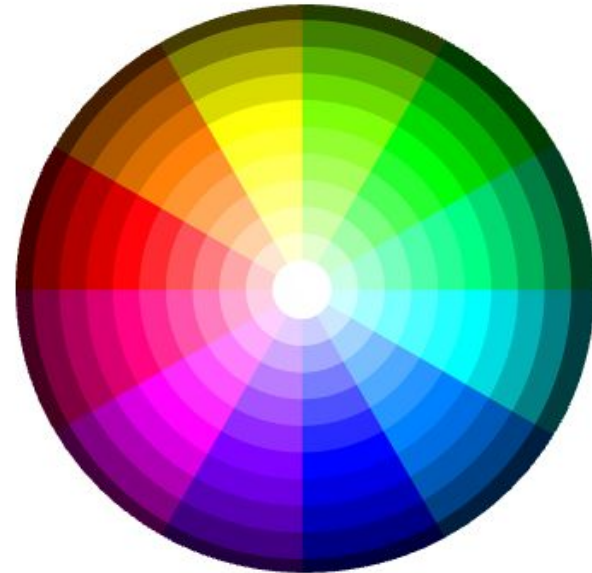
Q: Which colour best describes the one represented by the hexadecimal colour code: #32CD32?

- A. Shade of red
- B. Shade of blue
- C. Shade of green
- D. Shade of purple
- E. Shade of yellow



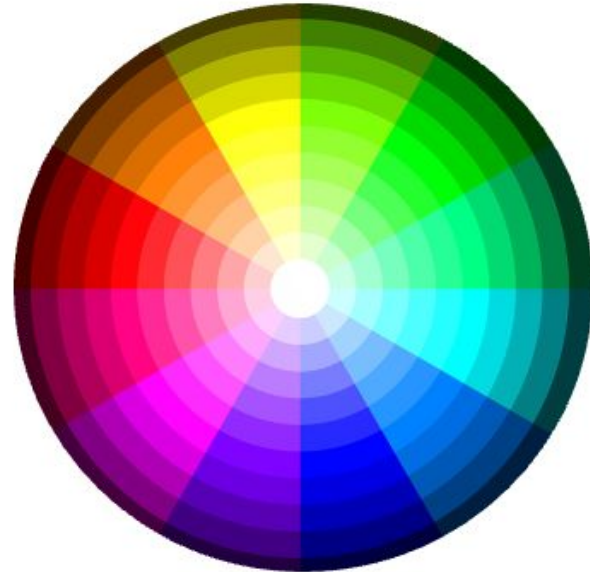
Q: Which colour best describes the one represented by the hexadecimal colour code: #32CD32?

- A. Shade of red
- B. Shade of blue
- C. Shade of green
- D. Shade of purple
- E. Shade of yellow



Q: Which colour best describes the one represented by the hexadecimal colour code: #800B80?

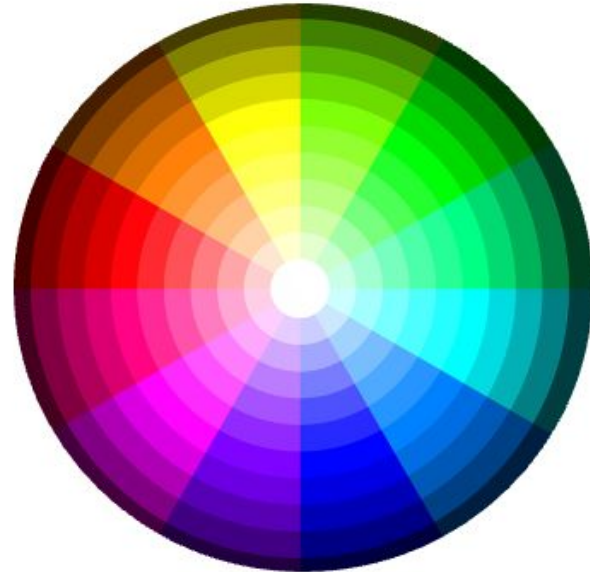
- A. Shade of red
- B. Shade of blue
- C. Shade of green
- D. Shade of purple
- E. Shade of yellow



Q: Which colour best describes the one represented by the hexadecimal colour code: #800B80?



- A. Shade of red
- B. Shade of blue
- C. Shade of green
- D. Shade of purple
- E. Shade of yellow



Activity



Color Mixing: Match the Colour to its Hex Rep.

Hex. Rep.	shade of ...
#FFA933	Yellow
#FF99FF	Pink or Magenta
#EAE51D	Blue
#A1A2A3	Orange
#1234F8	Grey



Color Mixing: Match the Colour to its Hex Rep.

Hex. Rep.		shade of ...
#FFA933		Yellow
#FF99FF		Pink or Magenta
#EAE51D		Blue
#A1A2A3		Orange
#1234F8		Grey



Color Mixing: Match the Colour to its Hex Rep.

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Color Mixing: Match the Colour to its Hex Rep.

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Optional Video about Colours

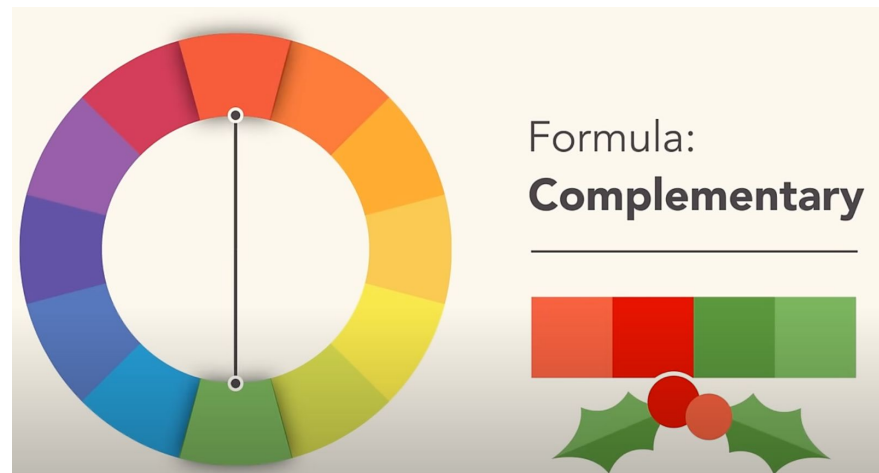
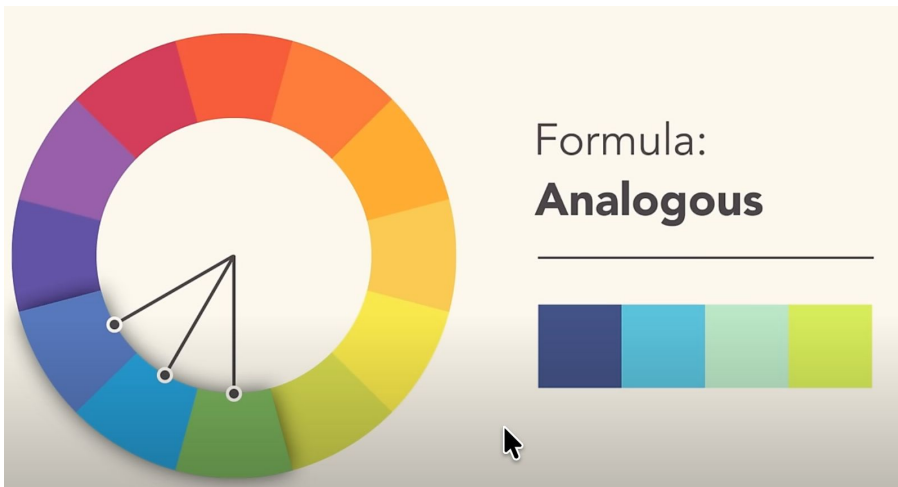
Colour Theory

https://youtu.be/_2LLXnUdUlc?si=ZC0gCVCkhlmnc3KT



Colour Theory

https://youtu.be/_2LLXnUdUlc?si=ZC0gCVCkhlmnc3KT



Data Representation in characters!

How do computer store letters and characters?



ASCII: Overview

American **S**tandard **C**ode for **I**nformation **I**nterchange (ASCII)

- Character encoding standard
- Allows computers and electronic devices to represent text
- Developed in 1960s
 - Standardize how computers represent characters

ASCII Table

Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char
0	0	[NULL]	32	20	[SPACE]	64	40	@	96	60	`
1	1	[START OF HEADING]	33	21	!	65	41	A	97	61	a
2	2	[START OF TEXT]	34	22	"	66	42	B	98	62	b
3	3	[END OF TEXT]	35	23	#	67	43	C	99	63	c
4	4	[END OF TRANSMISSION]	36	24	\$	68	44	D	100	64	d
5	5	[ENQUIRY]	37	25	%	69	45	E	101	65	e
6	6	[ACKNOWLEDGE]	38	26	&	70	46	F	102	66	f
7	7	[BELL]	39	27	'	71	47	G	103	67	g
8	8	[BACKSPACE]	40	28	(	72	48	H	104	68	h
9	9	[HORIZONTAL TAB]	41	29	)	73	49	I	105	69	i
10	A	[LINE FEED]	42	2A	*	74	4A	J	106	6A	j
11	B	[VERTICAL TAB]	43	2B	+	75	4B	K	107	6B	k
12	C	[FORM FEED]	44	2C	,	76	4C	L	108	6C	l
13	D	[CARRIAGE RETURN]	45	2D	-	77	4D	M	109	6D	m
14	E	[SHIFT OUT]	46	2E	.	78	4E	N	110	6E	n
15	F	[SHIFT IN]	47	2F	/	79	4F	O	111	6F	o
16	10	[DATA LINK ESCAPE]	48	30	0	80	50	P	112	70	p
17	11	[DEVICE CONTROL 1]	49	31	1	81	51	Q	113	71	q
18	12	[DEVICE CONTROL 2]	50	32	2	82	52	R	114	72	r
19	13	[DEVICE CONTROL 3]	51	33	3	83	53	S	115	73	s
20	14	[DEVICE CONTROL 4]	52	34	4	84	54	T	116	74	t
21	15	[NEGATIVE ACKNOWLEDGE]	53	35	5	85	55	U	117	75	u
22	16	[SYNCHRONOUS IDLE]	54	36	6	86	56	V	118	76	v
23	17	[ENG OF TRANS. BLOCK]	55	37	7	87	57	W	119	77	w
24	18	[CANCEL]	56	38	8	88	58	X	120	78	x
25	19	[END OF MEDIUM]	57	39	9	89	59	Y	121	79	y
26	1A	[SUBSTITUTE]	58	3A	:	90	5A	Z	122	7A	z
27	1B	[ESCAPE]	59	3B	;	91	5B	[	123	7B	{
28	1C	[FILE SEPARATOR]	60	3C	<	92	5C	\	124	7C	
29	1D	[GROUP SEPARATOR]	61	3D	=	93	5D	]	125	7D	}
30	1E	[RECORD SEPARATOR]	62	3E	>	94	5E	^	126	7E	~
31	1F	[UNIT SEPARATOR]	63	3F	?	95	5F	_	127	7F	[DEL]



ASCII: Overview

- Punctuation, spaces and other special control characters are also encoded
 - each encoded item is sometimes called a **code point**
- ASCII uses **7 bits** to represent each character, which allows for 128 (2^7) unique characters (from 0 to 127)
 - **Why 7 bits?**
 - An extra “check” bit to detect certain errors that might arise
- *Extended* ASCII uses **8 bits** (or one byte), allowing for characters with accents (Á, ë and others)



ASCII: Overview

ASCII uses **7 bits** to represent each character:

- **Control characters** (0–31): For managing hardware (like line breaks or bell sounds).
- **Printable characters** (32–126): Letters, digits, punctuation, and symbols.
- **Delete character** (127).

ASCII: Control characters (0-31)

Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char
0	0	[NULL]	32	20	[SPACE]	64	40	@	96	60	`
1	1	[START OF HEADING]	33	21	!	65	41	A	97	61	a
2	2	[START OF TEXT]	34	22	"	66	42	B	98	62	b
3	3	[END OF TEXT]	35	23	#	67	43	C	99	63	c
4	4	[END OF TRANSMISSION]	36	24	\$	68	44	D	100	64	d
5	5	[ENQUIRY]	37	25	%	69	45	E	101	65	e
6	6	[ACKNOWLEDGE]	38	26	&	70	46	F	102	66	f
7	7	[BELL]	39	27	'	71	47	G	103	67	g
8	8	[BACKSPACE]	40	28	(	72	48	H	104	68	h
9	9	[HORIZONTAL TAB]	41	29	)	73	49	I	105	69	i
10	A	[LINE FEED]	42	2A	*	74	4A	J	106	6A	j
11	B	[VERTICAL TAB]	43	2B	+	75	4B	K	107	6B	k
12	C	[FORM FEED]	44	2C	,	76	4C	L	108	6C	l
13	D	[CARRIAGE RETURN]	45	2D	-	77	4D	M	109	6D	m
14	E	[SHIFT OUT]	46	2E	.	78	4E	N	110	6E	n
15	F	[SHIFT IN]	47	2F	/	79	4F	O	111	6F	o
16	10	[DATA LINK ESCAPE]	48	30	0	80	50	P	112	70	p
17	11	[DEVICE CONTROL 1]	49	31	1	81	51	Q	113	71	q
18	12	[DEVICE CONTROL 2]	50	32	2	82	52	R	114	72	r
19	13	[DEVICE CONTROL 3]	51	33	3	83	53	S	115	73	s
20	14	[DEVICE CONTROL 4]	52	34	4	84	54	T	116	74	t
21	15	[NEGATIVE ACKNOWLEDGE]	53	35	5	85	55	U	117	75	u
22	16	[SYNCHRONOUS IDLE]	54	36	6	86	56	V	118	76	v
23	17	[ENG OF TRANS. BLOCK]	55	37	7	87	57	W	119	77	w
24	18	[CANCEL]	56	38	8	88	58	X	120	78	x
25	19	[END OF MEDIUM]	57	39	9	89	59	Y	121	79	y
26	1A	[SUBSTITUTE]	58	3A	:	90	5A	Z	122	7A	z
27	1B	[ESCAPE]	59	3B	;	91	5B	[	123	7B	{
28	1C	[FILE SEPARATOR]	60	3C	<	92	5C	\	124	7C	
29	1D	[GROUP SEPARATOR]	61	3D	=	93	5D	]	125	7D	}
30	1E	[RECORD SEPARATOR]	62	3E	>	94	5E	^	126	7E	~
31	1F	[UNIT SEPARATOR]	63	3F	?	95	5F	_	127	7F	[DEL]

ASCII: Printable characters (32-126)

Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char
0	0	[NULL]	32	20	[SPACE]	64	40	@	96	60	`
1	1	[START OF HEADING]	33	21	!	65	41	A	97	61	a
2	2	[START OF TEXT]	34	22	"	66	42	B	98	62	b
3	3	[END OF TEXT]	35	23	#	67	43	C	99	63	c
4	4	[END OF TRANSMISSION]	36	24	\$	68	44	D	100	64	d
5	5	[ENQUIRY]	37	25	%	69	45	E	101	65	e
6	6	[ACKNOWLEDGE]	38	26	&	70	46	F	102	66	f
7	7	[BELL]	39	27	'	71	47	G	103	67	g
8	8	[BACKSPACE]	40	28	(	72	48	H	104	68	h
9	9	[HORIZONTAL TAB]	41	29	)	73	49	I	105	69	i
10	A	[LINE FEED]	42	2A	*	74	4A	J	106	6A	j
11	B	[VERTICAL TAB]	43	2B	+	75	4B	K	107	6B	k
12	C	[FORM FEED]	44	2C	,	76	4C	L	108	6C	l
13	D	[CARRIAGE RETURN]	45	2D	-	77	4D	M	109	6D	m
14	E	[SHIFT OUT]	46	2E	.	78	4E	N	110	6E	n
15	F	[SHIFT IN]	47	2F	/	79	4F	O	111	6F	o
16	10	[DATA LINK ESCAPE]	48	30	0	80	50	P	112	70	p
17	11	[DEVICE CONTROL 1]	49	31	1	81	51	Q	113	71	q
18	12	[DEVICE CONTROL 2]	50	32	2	82	52	R	114	72	r
19	13	[DEVICE CONTROL 3]	51	33	3	83	53	S	115	73	s
20	14	[DEVICE CONTROL 4]	52	34	4	84	54	T	116	74	t
21	15	[NEGATIVE ACKNOWLEDGE]	53	35	5	85	55	U	117	75	u
22	16	[SYNCHRONOUS IDLE]	54	36	6	86	56	V	118	76	v
23	17	[ENG OF TRANS. BLOCK]	55	37	7	87	57	W	119	77	w
24	18	[CANCEL]	56	38	8	88	58	X	120	78	x
25	19	[END OF MEDIUM]	57	39	9	89	59	Y	121	79	y
26	1A	[SUBSTITUTE]	58	3A	:	90	5A	Z	122	7A	z
27	1B	[ESCAPE]	59	3B	;	91	5B	[	123	7B	{
28	1C	[FILE SEPARATOR]	60	3C	<	92	5C	\	124	7C	
29	1D	[GROUP SEPARATOR]	61	3D	=	93	5D	]	125	7D	}
30	1E	[RECORD SEPARATOR]	62	3E	>	94	5E	^	126	7E	~
31	1F	[UNIT SEPARATOR]	63	3F	?	95	5F	_	127	7F	[DEL]

ASCII: Delete character (127)

Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char
0	0	[NULL]	32	20	[SPACE]	64	40	@	96	60	`
1	1	[START OF HEADING]	33	21	!	65	41	A	97	61	a
2	2	[START OF TEXT]	34	22	"	66	42	B	98	62	b
3	3	[END OF TEXT]	35	23	#	67	43	C	99	63	c
4	4	[END OF TRANSMISSION]	36	24	\$	68	44	D	100	64	d
5	5	[ENQUIRY]	37	25	%	69	45	E	101	65	e
6	6	[ACKNOWLEDGE]	38	26	&	70	46	F	102	66	f
7	7	[BELL]	39	27	'	71	47	G	103	67	g
8	8	[BACKSPACE]	40	28	(	72	48	H	104	68	h
9	9	[HORIZONTAL TAB]	41	29	)	73	49	I	105	69	i
10	A	[LINE FEED]	42	2A	*	74	4A	J	106	6A	j
11	B	[VERTICAL TAB]	43	2B	+	75	4B	K	107	6B	k
12	C	[FORM FEED]	44	2C	,	76	4C	L	108	6C	l
13	D	[CARRIAGE RETURN]	45	2D	-	77	4D	M	109	6D	m
14	E	[SHIFT OUT]	46	2E	.	78	4E	N	110	6E	n
15	F	[SHIFT IN]	47	2F	/	79	4F	O	111	6F	o
16	10	[DATA LINK ESCAPE]	48	30	0	80	50	P	112	70	p
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18	12	[DEVICE CONTROL 2]	50	32	2	82	52	R	114	72	r
19	13	[DEVICE CONTROL 3]	51	33	3	83	53	S	115	73	s
20	14	[DEVICE CONTROL 4]	52	34	4	84	54	T	116	74	t
21	15	[NEGATIVE ACKNOWLEDGE]	53	35	5	85	55	U	117	75	u
22	16	[SYNCHRONOUS IDLE]	54	36	6	86	56	V	118	76	v
23	17	[ENG OF TRANS. BLOCK]	55	37	7	87	57	W	119	77	w
24	18	[CANCEL]	56	38	8	88	58	X	120	78	x
25	19	[END OF MEDIUM]	57	39	9	89	59	Y	121	79	y
26	1A	[SUBSTITUTE]	58	3A	:	90	5A	Z	122	7A	z
27	1B	[ESCAPE]	59	3B	;	91	5B	[	123	7B	{
28	1C	[FILE SEPARATOR]	60	3C	<	92	5C	\	124	7C	
29	1D	[GROUP SEPARATOR]	61	3D	=	93	5D	]	125	7D	}
30	1E	[RECORD SEPARATOR]	62	3E	>	94	5E	^	126	7E	~
31	1F	[UNIT SEPARATOR]	63	3F	?	95	5F	_	127	7F	[DEL]





Q: Convert **Hi! from ASCII to Decimal**



- A. 48, 69, 21
- B. 47, 68, 20
- C. 71, 104, 34
- D. 72, 105, 33
- E. None of the above



Q: Convert **Hi! from ASCII to Decimal**



A. 48, 69, 21

B. 47, 68, 20

C. 71, 104, 34

D. 72, 105, 33

E. None of the above



Q: Convert the following Hex to ASCII:

A. BADGE

B. *),/-

C. A@CFD

D.)(+.,

E. None of the above

Binary	01000010	01000001	01000100	01000111	01000101
Hex	42	41	44	47	45
ASCII					

Q: Convert the following Hex to ASCII:



A. BADGE

B. *) ,/-

C. A@CFD

D.)(+.,

E. None of the above

Binary	01000010	01000001	01000100	01000111	01000101
Hex	42	41	44	47	45
ASCII					

**Can we do this
faster?**



Convert Text to ASCII

Emoji ❤️ Red Heart on platforms



Apple



Google



Facebook



Twitter



Mozilla



Windows



Samsung



LG



Whatsapp



Openmoji



Telegram

Technical Information

Properties

Encoding

Copy & Paste Codes

Encoding	hex	dec (bytes)	dec	binary
UTF-8	E2 9D A4	226 157 164	14851492	11100010 10011101 10100100

Demo

Application of ASCII

The Martian

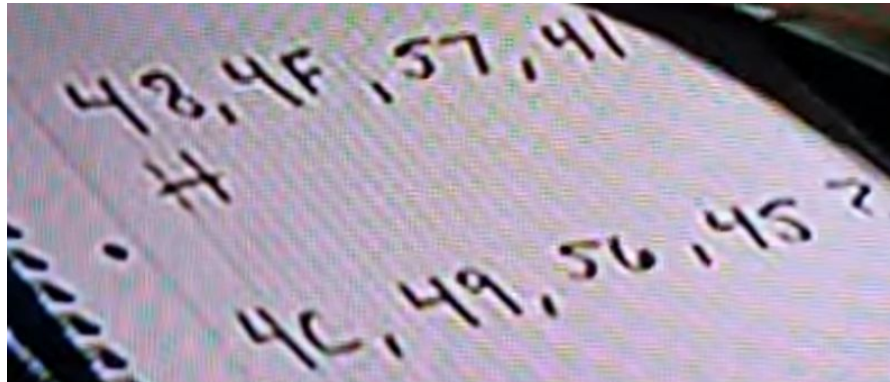
The Martian

- “Ridley Scott claimed that one of the most difficult scenes to direct was how to explain to the audience the hexadecimal system Watney uses as a code to communicate with Earth, which Scott admitted was hard for himself to understand.” [IMDb](#)
- Youtube Video:
 - [The Martian, Hexadecimal Scene](#)





The Martian



Decimal	Hex	Char
64	40	@
65	41	A
66	42	B
67	43	C
68	44	D
69	45	E
70	46	F
71	47	G
72	48	H
73	49	I
74	4A	J
75	4B	K
76	4C	L
77	4D	M
78	4E	N
79	4F	O
80	50	P
81	51	Q
82	52	R
83	53	S
84	54	T
85	55	U
86	56	V
87	57	W
88	58	X
89	59	Y
90	5A	Z
91	5B	[

What about other languages?

Arabic (ا-ي)

Chinese (汉字)

Emojis (😊, 🚀)

Unicode

Unicode



Unicode: Overview

Universal character **encoding** standard

- Designed to represent every character from every language in the world, as well as symbols, emojis, and special scripts, using a unified system.

Before unicode...

- Different encoding systems (like ASCII, ISO-8859-1, Shift-JIS)
 - Difficult to mix languages in one document
- Unicode solved this by creating **one global codebook**



How does Unicode Work?

- Unicode assigns a **unique number** called a “**code point**” to every character, regardless of platform, program, or language.
 - Code points are written like: U+0041 (which is ‘A’).
- Unicode itself is just a standard. To store the characters in files and transmit them over networks, you need **encoding formats**
 - There are different implementations, including UTF-8 and UTF-16 (UTF stands for **Unicode Transformation Format**)

Encoding Formats

- UTF-8 and UTF-16 are variable length encodings
- They use 1 byte (8 bits) for ASCII, but more for other characters

character	encoding	bits
A	UTF-8	01000001
A	UTF-16	00000000 01000001
A	UTF-32	00000000 00000000 00000000 01000001
あ	UTF-8	11100011 10000001 10000010
あ	UTF-16	00110000 01000010
あ	UTF-32	00000000 00000000 00110000 01000010

Emoji

- Notice how Emoji have the same “CLDR” names and Unicode values but different aesthetics

Smileys & Emotion								
face-smiling								
No	Code	Browser	Sample	GMail	SB	DCM	KDDI	CLDR Short Name
1	U+1F600				—	—	—	grinning face
2	U+1F603							grinning face with big eyes
3	U+1F604					—	—	grinning face with smiling eyes
4	U+1F601							beaming face with smiling eyes
5	U+1F606				—		—	grinning squinting face

Emoji

- This is how skin-tones are varied!

hand-fingers-open				
№	Code	Browser	Sample	CLDR Short Name
1	U+1F44B U+1F3FB			waving hand: light skin tone
2	U+1F44B U+1F3FC			waving hand: medium-light skin tone
3	U+1F44B U+1F3FD			waving hand: medium skin tone
4	U+1F44B U+1F3FE			waving hand: medium-dark skin tone
5	U+1F44B U+1F3FF			waving hand: dark skin tone

Source



Demo



How does Word store its data?

- Uploading a Word document into the online Hex editor suggests that the document is not in ASCII representation
- Most of the files that comprise a Word document are in **XML** (**Extensible Markup Language**) format; they describe metadata such as the font style and size, document creator, etc.
- The files may also contain information about tracked changes to the document, collaborators, privacy and security settings, and more



Activity

How does Microsoft Word store its data?

- Open Microsoft Word
- Write your name in the file and save it
- Visit <https://hexed.it/>
- Open the file using *Open file* feature
- What do you notice?

CPSC 100

Course Project



Class Activity

- Today we will make groups for the project!
- First, a bit more about the CPSC 100 course project and Contract Grading

Q: Which topic are you most interested in (not finalized) ?



- A. Generative AI and student learning
- B. Technological inequality
- C. Algorithmic Bias
- D. Financial and Environmental impacts of technology
- E. Technology and Crime

Participation Question

What grade do you intend to contract for?

A+

A

B

C



Take Home Practice

Q: Convert the following hexadecimal sequence to ASCII: 53 54 41 52 53



- A. STARS
- B. 56)45
- C. !@#\$\$\$
- D. HELLO
- E. WORLD

Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char
0	0	[NULL]	32	20	[SPACE]	64	40	@	96	60	`
1	1	[START OF HEADING]	33	21	!	65	41	A	97	61	a
2	2	[START OF TEXT]	34	22	"	66	42	B	98	62	b
3	3	[END OF TEXT]	35	23	#	67	43	C	99	63	c
4	4	[END OF TRANSMISSION]	36	24	\$	68	44	D	100	64	d
5	5	[ENQUIRY]	37	25	%	69	45	E	101	65	e
6	6	[ACKNOWLEDGE]	38	26	&	70	46	F	102	66	f
7	7	[BELL]	39	27	'	71	47	G	103	67	g
8	8	[BACKSPACE]	40	28	(	72	48	H	104	68	h
9	9	[HORIZONTAL TAB]	41	29	)	73	49	I	105	69	i
10	A	[LINE FEED]	42	2A	*	74	4A	J	106	6A	j
11	B	[VERTICAL TAB]	43	2B	+	75	4B	K	107	6B	k
12	C	[FORM FEED]	44	2C	,	76	4C	L	108	6C	l
13	D	[CARRIAGE RETURN]	45	2D	-	77	4D	M	109	6D	m
14	E	[SHIFT OUT]	46	2E	.	78	4E	N	110	6E	n
15	F	[SHIFT IN]	47	2F	/	79	4F	O	111	6F	o
16	10	[DATA LINK ESCAPE]	48	30	0	80	50	P	112	70	p
17	11	[DEVICE CONTROL 1]	49	31	1	81	51	Q	113	71	q
18	12	[DEVICE CONTROL 2]	50	32	2	82	52	R	114	72	r
19	13	[DEVICE CONTROL 3]	51	33	3	83	53	S	115	73	s
20	14	[DEVICE CONTROL 4]	52	34	4	84	54	T	116	74	t
21	15	[NEGATIVE ACKNOWLEDGE]	53	35	5	85	55	U	117	75	u
22	16	[SYNCHRONOUS IDLE]	54	36	6	86	56	V	118	76	v
23	17	[END OF TRANS. BLOCK]	55	37	7	87	57	W	119	77	w
24	18	[CANCEL]	56	38	8	88	58	X	120	78	x
25	19	[END OF MEDIUM]	57	39	9	89	59	Y	121	79	y
26	1A	[SUBSTITUTE]	58	3A	:	90	5A	Z	122	7A	z
27	1B	[ESCAPE]	59	3B	;	91	5B	[	123	7B	{
28	1C	[FILE SEPARATOR]	60	3C	<	92	5C	\	124	7C	
29	1D	[GROUP SEPARATOR]	61	3D	=	93	5D	]	125	7D	}
30	1E	[RECORD SEPARATOR]	62	3E	>	94	5E	^	126	7E	~
31	1F	[UNIT SEPARATOR]	63	3F	?	95	5F	_	127	7F	[DEL]

Q: Convert the following hexadecimal sequence to ASCII: 53 54 41 52 53



A. STARS

B. 56)45

C. !@#\$\$\$

D. HELLO

E. WORLD

Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char
0	0	[NULL]	32	20	[SPACE]	64	40	@	96	60	`
1	1	[START OF HEADING]	33	21	!	65	41	A	97	61	a
2	2	[START OF TEXT]	34	22	"	66	42	B	98	62	b
3	3	[END OF TEXT]	35	23	#	67	43	C	99	63	c
4	4	[END OF TRANSMISSION]	36	24	\$	68	44	D	100	64	d
5	5	[ENQUIRY]	37	25	%	69	45	E	101	65	e
6	6	[ACKNOWLEDGE]	38	26	&	70	46	F	102	66	f
7	7	[BELL]	39	27	'	71	47	G	103	67	g
8	8	[BACKSPACE]	40	28	(	72	48	H	104	68	h
9	9	[HORIZONTAL TAB]	41	29	)	73	49	I	105	69	i
10	A	[LINE FEED]	42	2A	*	74	4A	J	106	6A	j
11	B	[VERTICAL TAB]	43	2B	+	75	4B	K	107	6B	k
12	C	[FORM FEED]	44	2C	,	76	4C	L	108	6C	l
13	D	[CARRIAGE RETURN]	45	2D	-	77	4D	M	109	6D	m
14	E	[SHIFT OUT]	46	2E	.	78	4E	N	110	6E	n
15	F	[SHIFT IN]	47	2F	/	79	4F	O	111	6F	o
16	10	[DATA LINK ESCAPE]	48	30	0	80	50	P	112	70	p
17	11	[DEVICE CONTROL 1]	49	31	1	81	51	Q	113	71	q
18	12	[DEVICE CONTROL 2]	50	32	2	82	52	R	114	72	r
19	13	[DEVICE CONTROL 3]	51	33	3	83	53	S	115	73	s
20	14	[DEVICE CONTROL 4]	52	34	4	84	54	T	116	74	t
21	15	[NEGATIVE ACKNOWLEDGE]	53	35	5	85	55	U	117	75	u
22	16	[SYNCHRONOUS IDLE]	54	36	6	86	56	V	118	76	v
23	17	[END OF TRANS. BLOCK]	55	37	7	87	57	W	119	77	w
24	18	[CANCEL]	56	38	8	88	58	X	120	78	x
25	19	[END OF MEDIUM]	57	39	9	89	59	Y	121	79	y
26	1A	[SUBSTITUTE]	58	3A	:	90	5A	Z	122	7A	z
27	1B	[ESCAPE]	59	3B	;	91	5B	[	123	7B	{
28	1C	[FILE SEPARATOR]	60	3C	<	92	5C	\	124	7C	
29	1D	[GROUP SEPARATOR]	61	3D	=	93	5D	]	125	7D	}
30	1E	[RECORD SEPARATOR]	62	3E	>	94	5E	^	126	7E	~
31	1F	[UNIT SEPARATOR]	63	3F	?	95	5F	_	127	7F	[DEL]

Convert your name into its ASCII values

Convert your name into its Hex format



Convert your name into its Binary format

Use <https://symbl.cc/> to find the unicode for the following characters:

